

GURSHARAN SINGH

Full-Stack Software Engineer • React • Next.js • Node.js • TypeScript

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TECHNICAL SKILLS

Frontend: React, Next.js (App Router), TypeScript, JavaScript (ES6+), Tailwind CSS, HTML5, CSS3
Backend: Node.js, Express.js, RESTful APIs, JWT Authentication, Zod, Python (Automation)
Databases: MongoDB (Mongoose), PostgreSQL, Database Schema Design, Compound Indexing
Tools & Workflow: Git, GitHub, Postman, Vercel, Vite, Chrome DevTools, Linux/CLI

PROFESSIONAL EXPERIENCE

Full-Stack Web Developer — Freelance / Independent Engineering

Jan 2024 – Present

- Engineered and shipped responsive client web apps using **React**, **Next.js**, and **Tailwind CSS**, focusing on fast mobile load times and clean component architecture.
- Developed modular **Node.js & Express** REST APIs with structured endpoints for user onboarding, profile management, and contact submissions.
- Integrated external third-party services including **Stripe Checkout** for secure customer billing and **Nodemailer** for automated transactional notifications.
- Collaborated directly with early-stage business owners to scope technical requirements and delivered working functional MVPs within **24–48 hours** to validate product roadmaps.

Frontend Web Developer — TechSolutions Digital Studio

Jun 2023 – Dec 2023

- Translated Figma and Adobe XD prototypes into clean, responsive React interfaces with strict adherence to design system specifications and mobile-first layouts.
- Managed client-side application state using **React Context API** and custom hooks, eliminating prop-drilling across multi-step user onboarding flows.
- Connected frontend components to backend REST APIs, implementing intuitive skeleton loaders, error boundaries, and form validation to maximize usability.
- Coordinated code versioning, feature branches, and pull request reviews across team members using **Git and GitHub**.

FEATURED ENGINEERING PROJECTS

DineFlow 2.0 — Hospitality OS & KDS [\[JavaScript ES6+, BroadcastChannel, Web Audio API, POS Receipts\]](#) [\[Live\]](#) Full-Stack Platform
[Demo ↗](#) [\[Code →\]](#)

- Engineered real-time restaurant OS with dedicated Guest Dining, Kitchen KDS, and Floor POS workspaces synchronized via **BroadcastChannel** with sub-5ms cross-window latency.
- Built pure mathematical Web Audio synthesizer replicating acoustic brass counter bell (1760Hz) with 0ms lag, and generated 80mm dot-matrix thermal tax receipts.

Playify — Music Streaming PWA & Android App [\[PWA, WebRTC PeerJS, MQTT, DES Crypto, Service Worker\]](#) [\[Live\]](#) Media & Systems
[Demo ↗](#) [\[Code →\]](#)

- Constructed an offline-first music streaming PWA featuring real-time multi-device playback synchronization (<10ms clock phase-lock via WebRTC DataChannels & MQTT).
- Packaged standalone Android APK (1.5MB) with native MediaSession integration, background playback, and client-side DES audio stream decryption.

PowerSync — Deterministic Offline-First Sync Backend [\[Node.js, Express, PostgreSQL, SQLite, RS256 JWT, JWKS\]](#) Distributed Systems
[\[Code →\]](#)

- Architected bidirectional sync backend enabling offline-first client replication between edge SQLite and server-side PostgreSQL.
- Implemented asymmetric RS256 JWT token issuance and exposed standard JWKS endpoint for automated replica authentication.

ScoutFlow — Autonomous Market Intelligence Daemon [\[Python 3, BeautifulSoup, Telegram Bot API, GitHub Actions\]](#) [\[Code →\]](#) Cloud Automation

- Engineered a 24/7 background pipeline scraping 8+ remote technical feeds with SHA-256 state deduplication.
- Extracted verified direct founder contact addresses and dispatched real-time alerts via Telegram Bot webhook.